

Economic Impact of French Revolution

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Abstract

In this paper, we revisit, replicate, and critically appraise the work of [Acemoglu et al., 2011] on the economic impact of the French Revolution on neighboring territories. The grander objective of the researchers here is to investigate the consequences of radical, externally imposed reforms on subsequent economic growth. By using urbanization rates as a proxy for economic development, the researchers employ a least squares regression on a collection of interactions between French presence and the year dummies to test their hypotheses.

1 Summary

The French Revolution and France's occupation of Germany called for an extensive and radical set of reforms including the implementation of the civil legal code, abolition of guilds and feudalism, the introduction of equality before law, end of the French monarchy, and the undermining of French aristocracy. These institutional reforms are interesting both historically speaking as well as in current debates on the economic prosperity of nations with externally imposed institutions and law. However, the long-run implications of the radical changes stemming from the French Revolution and Napoleon's rule remain unclear. The two main conflicting views are: (1) externally imposed institutions like those designed by the French in the 1700s are unlikely to encourage economic progress. (2) oligarchies, entry barriers, and restrictions on trade in labor and other markets which were eradicated during the French Revolution would foster rapid economic growth in the areas that the French affected.

The researchers attempt to gauge the French revolution's long-run consequences on Germany's economy, considering the radical institutional reforms imposed on neighboring countries as a quasi-natural experiment. In particular, parts of Germany, mostly from the west and northwest, were invaded, ruled, and reformed by France while most of the south and southwest parts of Germany remained unaffected. The west and northwest regions were targeted more so than the south of Germany primarily due to competition - it was a strategic ploy to save resources and maintain authority. Prussia, Austria, and several other major governments in close proximity to the southern half of Germany would impede and exhaust the French's resources had they tried invading there. Now, to estimate the effects of French institutional reform in Europe, the French treatment is defined as the number of years between 1792 and 1815 that the German polity in question was under French control and they use urbanization rates as the main measure for economic prosperity.

In (1), they found no evidence that areas under French occupation experienced less rapid development. On the contrary, all of their evidence, including greater urbanization growth rates, points to greater rapid economic growth as proxied by urbanization in areas that underwent the radical institutional reforms brought by the French Revolution, especially after 1850. The results are inconsistent

with the view that externally imposed, radical, and “Big Bang” style reforms impede economic growth and is more in line with the view that the eradication of oligarchies, entry barriers, and restrictions on trade in labor and other markets encourage rapid economic growth.

The primary data set considered in (1) is found here. The data sets consist of 19 distinct preunitary polities in Germany with features described in table 3 in the appendix. It should be noted that two features, distance to Paris and share of the population of Protestant faith around 1800, were inferred by the authors themselves. They also employ data on the sectoral composition of employment for 64 German districts for the years 1849, 1882, 1895, and 1907 (from Harald Frank 1994) that serves as a proxy for industrialization in the polities. Table 1 lists all 19 polities in the data set sorted by their treatment status including how many years each polity was under French presence, civil code adoption, abolition of serfdom and guilds, agrarian reforms, and an index of reforms in 1850 and 1900, and population weights. The index quantifies the distinction between different German polities that were reformed by the French. The averages calculated are weighted based on the population weights. Table 2 goes over the weighted sample mean and variance (weighted by total population in 1750) of urbanization, share Protestant, latitude, longitude, and distance to Paris for both samples of 13 polities west of the Elbe (five of them in the treatment area) and the 19 polities. Differences in urbanization rates before 1800 were very small between the two groups; however, by 1900 urbanization rates West of Elbe were almost 8% higher than areas occupied by Napoleon.

Country	Years of French Presence	Civil Code	Abolition of serfdom	Agrarian reform	Abolition of guilds	Reforms index as of 1850	Reforms index as of 1900	Pop. weights
Rhineland	19	1802	1798	1804	1795	50.25	100.25	1439
Bavarian Palatinate	19	1802	1798	1804	1795	50.25	100.25	239
Mark	6	1810-15, 1900	1808	1825	1809	28.25	65.75	150
Westphalia without Mark	6	1810-15, 1900	1808	1825	1809	28.25	65.75	529
Brunswick	6	1808-14, 1900	1808-18, 1834	1809-18, 1834	1808-15, 1864	16.00	50.00	155
Saxony Province	6	1808-15, 1900	1808	1809	1809	32.75	70.25	763
Hessen-Kassel	6	1808-14, 1900	1808-14, 1832	1809-14, 1832	1808-16, 1869	15.25	48.00	294
Hanover	3	1808-13, 1900	1808-14, 1833	1809-14, 1833	1808-15, 1869	14.25	47.00	1090
Average	9.98					32.41	72.88	
Baden	0	1810	1783	1820	1862	34.25	81.25	609
Bavaria Old	0	1900	1808	1826	1868	16.50	49.50	1163
Hessen-Darmstadt, right of the Rhine	0	1900	1811	1816	1866	18.25	51.75	264
Saxony	0	1865	1832	1832	1862	9.00	52.25	1020
Wuerttemberg	0	1900	1817	1836	1862	11.75	46.25	925
Average	0					16.31	54.46	
Brandenburg	0	1900	1811	1821	1810	27.0	64.50	797
East Prussia	0	1900	1811	1821	1810	27.0	64.50	554
Pomerania without Swedish Pomerania	0	1900	1811	1821	1810	27.0	64.50	342
Silesia	0	1900	1811	1821	1810	27.0	64.50	1053
Mecklenburg-Schwerin	0	1900	1820	1862	1869	7.5	37.25	217
Schleswig-Holstein	0	1900	1805	1805	1867	22.5	55.75	541
Average	0					25.1	61.46	

Table 1: Replication of Table 1

2 Replication

Here, we replicate the main results of the paper that are listed in table 3 of (1) (our table 3).

	West of Elbe All	West of Elbe Control	Treatment	Whole Sample Control	Whole Sample All
Urbanization in 1700	5.86	5.49	6.35	7.68	7.27
(*)	4.22	2.89	5.93	5.69	5.57
Urbanization in 1750	7.89	7.66	8.09	8.9	8.59
(*)	4.12	4.19	4.4	4.91	4.61
Urbanization in 1800	10.23	8.45	11.75	10.22	10.81
(*)	4.37	3.47	4.71	5.16	4.92
Urbanization in 1850	13.78	12.38	14.96	14.45	14.65
(*)	6.01	5.42	6.65	7.9	7.26
Urbanization in 1875	25.7	22.22	28.67	25.19	26.52
(*)	10.7	9.15	11.69	11.2	11.21
Urbanization in 1900	39.89	35.6	43.55	38.27	40.29
(*)	13.92	12.12	15.2	14.4	14.54
Share Protestant	0.58	0.53	0.62	0.68	0.65
(*)	0.36	0.42	0.32	0.36	0.34
Latitude	50.43	49.28	51.41	50.97	51.14
(*)	1.55	1.26	1.03	2.31	1.89
Longitude	9.89	10.89	9.03	12.78	11.34
(*)	2.08	2.28	1.59	3.63	3.5
Distance to Paris	592.38	631.82	558.66	808.86	712.88
(*)	151.29	173.69	135.61	272.06	256.75

Table 2: Replication of Table 2 - (*) denotes sample variance

2.1 Statistical Assumptions

For the main results, (1) makes the following assumptions

- The urbanization rate is a linear function of the interaction terms (the time effects $\{d_\tau\}$ times the number of years of French presence $\{I_j\}$), a set of time effects, and a set of polity fixed effects $\{\delta_j\}$:

$$u_{jt} = d_t + \delta_j + \sum_{\tau \in \mathcal{T}^{pre}} \alpha_\tau \times d_\tau \times I_j + \sum_{\tau \in \mathcal{T}^{post}} \alpha_\tau \times d_\tau \times I_j + \mathbf{X}_{jt}^T \times \gamma + \varepsilon_{jt}$$

u_{jt} is the urbanization rate for polity j at time t . The d_t s denotes the full set of time effects and the δ_j s are the full set of polity fixed effects. $\mathbf{X}_{jt}$ is a vector of other covariates included in the robustness checks. ε_{jt} is the error/noise term for polity j at time t . I_j , the key variable of interest, is the number of years of French presence. The coefficients of interest are $\{\alpha_t\}_{\tau \in \mathcal{T}^{pre}}$ and $\{\alpha_t\}_{\tau \in \mathcal{T}^{post}}$; $\mathcal{T}^{pre}$ is the set of years before the French treatment and $\mathcal{T}^{post}$ is the set of years after the French treatment. Separating the summation as above emphasizes the pre and post-French Revolution differential effects.

- The regression model errors are assumed to be independent across clusters (polities) but correlated within clusters. Each of the error terms for each cluster, ϵ_j are assumed to be independent and normal for each polity j , where each polity has an error covariance matrix.
- The data matrix is assumed to be nonsingular; if not, then the covariate(s) in question are dropped.
- Notice that in table 3, columns 1 and 3, a weighted regression is used based on the population of each polity. We assume the general covariance structure of the error term for each polity j is

additionally determined by the inverse of the total population for each polity in 1750. Columns 2 and 4 are not weighted regressions.

2.2 Regression Replication

	Dependent Variable: urbanization rate			
	West of Elbe Weighted	West of Elbe Unweighted	All Weighted	All Unweighted
Years French presence $\times$ 1750	-0.491	-0.252	-0.488	-0.197
(*)	0.249	0.172	0.235	0.164
Years French presence $\times$ 1800	-0.247	-0.043	-0.268	-0.047
(*)	0.225	0.153	0.227	0.178
Years French presence $\times$ 1850	-0.16	0.033	-0.221	-0.023
(*)	0.25	0.153	0.249	0.181
Years French presence $\times$ 1875	0.402	0.354	0.266	0.252
(*)	0.326	0.295	0.303	0.299
Years French presence $\times$ 1900	0.634	0.529	0.503	0.506
(*)	0.408	0.401	0.376	0.423
Observations	74	74	108	108
Number of states	13	13	19	19
p-value for joint significance > 1800	0.053	0.463	0.02	0.214

Table 3: Replication of Table 3 - (*) denotes sample variance

There is no change in the interpretation of the parameters since the estimated parameters are algebraically identical between the linear regression model with heteroskedasticity and the transformed OLS model (which gives the WLS estimator). Thus, all coefficients in this table, which are interaction terms, are interpreted similarly (the only change is in the dummy variable year in question). Let us consider interpreting the first coefficient in table 3, column 1, $\hat{\alpha}_{1750} = -0.491$. This means that, all else being held constant, a 1 year increase in French presence for the year 1750 versus not being in the year 1750 is associated with a 0.491 decrease in the urbanization rate on average.

Recall from the model that $\mathcal{T}^{post}$ includes the years 1850, 1875, and 1900, while $\mathcal{T}^{pre}$ includes the years 1750 and 1800. In column 1, we observe $\hat{\alpha}_{1750} = -0.491$ and $\hat{\alpha}_{1800} = -0.247$, which indicate that areas with larger years of French presence grew somewhat slowly in the 1700s, but this differential declines in 1800. Furthermore, $\hat{\alpha}_{1850} = -0.160$ with a corresponding standard error of 0.250, which implies that by 1850, there was no statistically significant association for the years of French presence on the growth of urbanization. By 1875 ($\hat{\alpha}_{1875} = 0.402$), there is a positive association between the years of French presence and urbanization rates, and by 1900 ($\hat{\alpha}_{1900} = 0.634$), this association strengthens.

For the F-test for the joint significance of all post-treatment years, we test the following hypothesis: $H_0 : \alpha_{1850} = \alpha_{1875} = \alpha_{1900} = 0$. The researchers state that since the p-value (0.0532) of the F-test is statistically significant at the 5% percent level of significance, we can reject the H_0 and there is evidence to suggest the post-treatment years are jointly statistically different than the pre-1850 dates. However, one should be wary interpreting this result because the p-value is slightly above the 5% significance level threshold.

The second column repeats the same analysis but minus the population weights in 1750. In this case, the results remain qualitatively similar since we observe a negative trend before the treatment and a positive trend afterward. However, the F-test no longer results in a statistically significant p-value. This means that we cannot reject the above H_0 and the post-treatment years are not jointly statistically different from the pre-treatment years.

In the third column, the coefficients have similar magnitudes compared to the first column and the F-test for the joint significance test results in a p-value of about 0.0205. The fourth column repeats the regression in the third column again but without the population weights; qualitatively, the fourth column matches the third column, but the F-test no longer yields in a p-value that is statistically significant.

The regressions from table 3 indicate that there is no evidence to suggest that there was a negative effect of French presence on the German polities and that there is some evidence to suggest that the polities in question experienced more rapid urbanization, particularly after 1850. The researchers indicate that the non-positive short-term effects stem from lingering exploitative and destructive aspects of the war.

2.3 Assumption Appraisal

In this section, we will critically appraise their statistical assumptions for the linear models they implemented in their study.

One statistical assumption lies in the choice of weights when implementing the weighted least squares regression in column 1 and 3. The researchers justify that unweighted regressions can be affected by sudden changes in urbanization in smaller polities. Thus, weighted regressions give a greater weight to changes in larger polities. However, the choice to use the population in 1750 as the weights is not well-justified and arbitrary - during this particular span of years, the revolution and other ongoing international events would have certainly catalyzed migrations and population shifts so the weights could not be classified as stationary.

The model presented in the paper assumes there exists a linear relationship between the independent variables and the dependent variable. However, urbanization rates often grow exponentially over time - below, we observe that the relationship between time and urbanization rates for the given data may in fact be nonlinear.

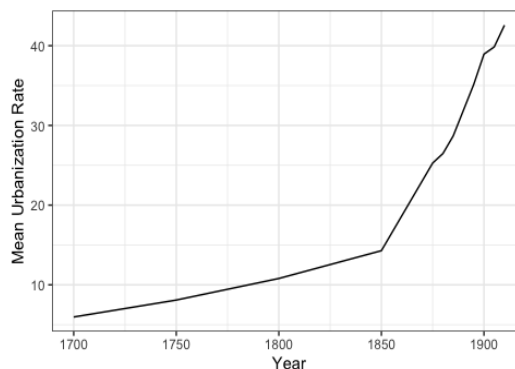


Figure 1: Year versus the Mean Urbanization Rate

Furthermore, the researchers assume independence among the urbanization rates for different polities. This is a challenging assumption to satisfy because these polities are located in Germany and thus more likely to experience spillover effects in the facilitation of trade and cross-communication. For instance, if polity A shares a border with polity B that has high urbanization rates, then polity A may eventually experience a spillover effect and observe high urbanization rates.

While implementing weighted and unweighted regressions, the data matrix was singular when we included the time effect for the year 1900. Thus, to invert the matrix $X^T X$ (where X is the data matrix), we drop this redundant variable(s) *yr1900*. As a result, this assumption is trivially satisfied.

3 Robustness Checks

The robustness checks presented in table 4 examine how the regression coefficient estimates $\hat{\alpha}_t$ (where $t \in \{1750, 1800, 1850, 1875, 1900\}$) behave when the regression specification is modified by including regressors and removing the polity Mark.

Table 4 includes the robustness checks of the basic results found in table 3 column 1. Column 1 drops the polity Mark (leave-one-cluster-out) as the presence of coal in the region may have sparked a differential growth advantage in the latter half of the 19th century. Columns 2-6 include a set of interactions between the time-effects and time-invariant features that could cause different urbanization development. For instance, in column 2, we include the interaction of the time effects and proportion of the Protestant population in the year; we can represent this as $\sum_{\tau} \eta_{\tau} \times D_{\tau} \times Protestant_j$. The researchers also use the interactions between the year dummies and latitude (column 3) and longitude (column 4) to address the potential time-varying effects of geography. Column 5 addresses the interaction with time effects and the distance to Paris. In particular, the researchers argue that the fourth and fifth columns are more important because areas that are further west and closer to Paris are more likely to be occupied by French forces. Additionally, some polities, such as Rhineland and Württemberg, resulted from the dissolution and merger of minor territories. As a result, column 6 addresses the possibility that the removal of internal borders contributes towards differential growth by adding the set of interaction terms between the year dummies and the number of old territories in that year. Furthermore, in column 7, the researchers address the concern that the initial urbanization (urbanization in 1750) may affect development growth due to mean-reverting dynamics in the growth of different polities. Consequently, they include a full set of interactions between the year dummies and initial urbanization.

The coefficient estimates in columns 1 to 7 seem to be similar to the baseline coefficient estimates with a negative pretrend and a positive association by 1875. The coefficients of interest $\{\alpha_t\}_{t \in \mathcal{T}^{pre}} \alpha_{\tau}$ are negative before the French treatment and then positive after the French treatment. In column 4, the researchers claim that the positive effects in 1875 and 1900 are both larger and are more precise (have lower standard errors). This reassures that there is a positive association by the year 1875. As for the p-values in the 8th row, we see that none of the results are significant now at a 5% level - implying that we accept the null that the interaction coefficients post-1800 are zero (contradicting the researcher's main results). For the p-values in column 2-5 and 7, they are testing the hypothesis: $H_0 : \eta_{1750} = \eta_{1800} = \eta_{1850} = \eta_{1875} = \eta_{1900} = 0$, where η_t corresponds to the coefficient of the interaction term between the year t dummy and the variable in question. For the p-value in column 6 (number of old territories), they are testing for $H_0 : \eta_{1850} = \eta_{1875} = \eta_{1900} = 0$ where η_t refers to

	Dependent Variable: urbanization rate						
	Without Ruhr	Protestant	Latitude	Longitude	Distance to Paris	#(old territories)	Initial urbanization
Years French presence x 1750	-0.565	-0.547	-0.6	-0.435	-0.446	-0.531	-0.458
(*)	0.281	0.281	0.356	0.244	0.278	0.294	0.267
Years French presence x 1800	-0.324	-0.336	-0.356	-0.256	-0.244	-0.288	-0.209
(*)	0.249	0.249	0.324	0.215	0.248	0.264	0.245
Years French presence x 1850	-0.235	-0.229	-0.261	0.018	-0.027	-0.114	-0.136
(*)	0.275	0.275	0.356	0.21	0.242	0.257	0.261
Years French presence x 1875	0.324	0.353	0.292	0.623	0.561	0.374	0.418
(*)	0.324	0.324	0.374	0.25	0.271	0.305	0.284
Years French presence x 1900	0.552	0.573	0.512	0.893	0.836	0.656	0.65
(*)	0.411	0.411	0.451	0.321	0.325	0.382	0.363
Observations	68	74	74	74	74	74	74
Number of states	12	13	13	13	13	13	13
p-value for joint significance > 1800	0.128	0.071	0.176	0.079	0.063	0.096	0.109
p-value for joint significant covariates		0.022	0.414	0.022	0.081	0.027	0.123

Table 4: Replication of Table 4 - (*) denotes sample variance

the coefficient for the interaction term for the dummy year t and number of old territories. Only columns 2, 4, and 6 yield p-values that are statistically significant at the 5% level, so we may reject the corresponding H_0 for these columns.

The researchers use the urbanization rate of the area as their main measure of economic prosperity, but one may argue that the urbanization rate is an insufficient measure of economic prosperity. Therefore, table 5 considers the possibility of using the sectoral composition of employment in agriculture and in industry, where more rapid industrialization indicates economic prosperity. Table 5 shows that in the West of Elbe, the share of population in agriculture is negative and transitioning towards a more industrial society by 1849; however, these differences are not statistically significant. By 1882, these values are significant and larger in magnitude, which indicate that there is more rapid industrialization in areas with French occupation.

	Dependent Variable: agriculture employment			Dependent Variable: industry employment		
	West of Elbe		All	West of Elbe		All
	Weighted	Unweighted	Weighted	Weighted	Unweighted	Weighted
Years French presence, 1849	-0.43	-0.411	-0.508	0.055	0.061	0.57
(*)	0.468	0.46	0.346	0.376	0.342	0.237
Years French presence, 1882	-0.45	-0.486	-0.585	0.42	0.386	0.57
(*)	0.285	0.244	0.253	0.256	0.24	0.237
Years French presence, 1895	-0.57	-0.601	-0.658	0.472	0.449	0.57
(*)	0.266	0.242	0.182	0.248	0.231	0.237
Years French presence, 1907	-0.554	-0.585	-0.724	0.35	0.321	0.57
(*)	0.281	0.264	0.237	0.284	0.251	0.237

Table 5: Replication of Table 5 - (*) denotes sample variance

4 Re-Analysis

In this section, we will compare and contrast our findings with the main result and provide justification for our reanalysis method of choice. Similar to the robustness check in table 4 column 1, we implement

leave-one-cluster-out for all of the West of Elbe polities. Table 6 presents the leave-one-polity-out results that yielded a statistically significant p-value (testing for $H_0 : \alpha_{1850} = \alpha_{1875} = \alpha_{1900} = 0$) at the 5% threshold.

	Dependent Variable: urbanization rate			
	Bavarian Palatinate	Saxony Province	Saxony	Hessen-Kassel
Years French presence $\times$ 1750	-0.718	-0.416	-0.341	-0.448
(*)	0.574	0.211	0.145	0.227
Years French presence $\times$ 1800	-0.452	-0.173	-0.12	-0.202
(*)	0.564	0.193	0.133	0.209
Years French presence $\times$ 1850	-0.372	-0.088	0.028	-0.115
(*)	0.595	0.214	0.121	0.234
Years French presence $\times$ 1875	0.29	0.481	0.695	0.45
(*)	0.683	0.313	0.171	0.323
Years French presence $\times$ 1900	0.548	0.716	1.004	0.685
(*)	0.757	0.392	0.199	0.402
Observations	68	68	68	68
Number of states	12	12	12	12
p-value for joint significance after 1800	0	0.024	0.001	0.03

Table 6: Re-analysis (Leave-one-cluster-out) - (*) denotes sample variance

Instead of using the entire dataset to fit the linear model, we will omit one polity from the model and see if the coefficients are significantly different from each other. The idea is that, although we have a relatively small dataset of 19 polities, removing one observation should not greatly affect the coefficients. In other words, no one observation should greatly change the main results that we found in table 3, column 1. We observe from table 7 and 8 that there is generally a negative pretrend in terms of urbanization rates and a positive association by 1875. When we remove Rhineland (table 8, column 1) from our regression, the 1875 interaction coefficient is negative but not significant. Furthermore, when we remove the polities in table 8, we observe that the p-values for the joint significance after 1800 are greater than our 5% threshold. The polities left out with nonsignificant p-values are close to Paris. The polities that are close to Paris are likely to be influenced by French reforms and, if they experience high urbanization rates, then removing this polity will greatly change the p-value to be greater than the 5% threshold. As a result, this aligns with (and strengthens) Acemoglu’s findings.

In this section, we perform leave-one-cluster-out analysis with the aim of robustifying Acemoglu’s results. Table 6 and table 8 (in the appendix) include the results of this estimation leaving each polity from the West of Elbe out once. The results mostly align with the central hypothesis of (1) that proximity to Paris is inversely proportional to the influence of the French regime and moreover economic prosperity with two notable exceptions. One exception is the Bavarian Palatinate which is one of the closest polities to Paris but, leaving it out yields jointly significant interaction terms post 1800. One hypothesis for this rather odd result is that the Bavarian Palatinate was disassembled in 1803 and from then on, its territory was continually updated and ruled by neighboring governments such as Austria and Prussia, that inhibited the urbanization growth experienced by more uniformly

(or consistently) ruled German polities out west. Thus, while the territory was strongly influenced by French rule during Napoleon's time, the hope of economic prosperity was obscured and tarnished by ongoing wars, takeovers, and the like. Another notable exception is the removal of the Hesse-Kassel polity, which likely led to a lower p-value because most French reforms made here were reverted by the old government once the French lost control. As a result, one may postulate that the Hesse-Kassel region did not experience the French treatment to the same degree as other polities and thus lower urbanization rates.

5 Conclusion

Ultimately, this replication study reaffirms Acemoglu's findings that there is no evidence the French-imposed reforms had negative economic consequences proxied by urbanization rates; in fact, we observe positive economic development. Our reanalysis demonstrates a similar negative pretrend and positive association by 1875, which further reconfirms the researchers' hypothesis. While there are anomalies in this analysis that contradict their findings, this may be largely due to other factors not reflected in the data but in history itself.

One major setback of this study is that the analysis is restricted to historical data and estimates of the urbanization rate and the independent variables. Furthermore, since the analysis is restricted to the French Revolution, we may not be able to extrapolate these results to other eras and regions. Future papers that wish to reconfirm the researchers' findings may want to use a different measurement of French "treatment" and examine if the results under a different treatment for institutional reform indicate stronger economic development. Nonetheless, their findings and our reconfirmation of these findings suggest that radical institutional reforms can result in long-run beneficial consequences. Our code for the replication can be found [here](#).

References

- [1] D. Acemoglu, D. Cantoni, S. Johnson, and J. Robinson, “The consequences of radical reform: The french revolution,” *American Economic Review*, vol. 101, pp. 3286–3307, 2011.

6 Appendix

Data Description	
Variable	Description and Source
Distance to Paris	Great circle distance from a territory’s capital to Paris—own calculation, based on latitude and longitude data.
Latitude	Latitude in degrees of a territory’s capital—Wikipedia.
Longitude	Longitude in degrees of a territory’s capital—Wikipedia.
Number of prerevolutionary territories	Number of territories of the Holy Roman Empire (immediate to the Emperor) existing prior to the invasion of French Revolutionary armies and the dissolution of the Empire through the Reichsdeputationshauptschluss—see online Appendix C .
Share agriculture	Share of total population (in percent) whose primary employment is in the agricultural sector—Frank (1994)
Share industry	Share of total population (in percent) whose primary employment is in the industrial and manufacturing sector—Frank (1994)
Share Protestant	Approximate share of the population of Protestant (Lutheran/ Calvinist) faith around 1800—own projection, based on nineteenth century census data.
Urbanization rate	Percent of total population living in cities above 5,000 inhabitants— see the volumes of Statistik des deutschen Reiches.
Years French presence	Years of presence of French troops or rule by French-appointed rulers—Grab (2003).

Table 7: Data Description

	Dependent Variable: urbanization rate								
	Rhineland	Mark	Westphalia	Brunswick	Hanover	Wuerttemberg	Baden	Bavaria Old	Hessen-Darmstadt
Years French presence $\times$ 1750	-0.103	-0.565	-0.492	-0.492	-0.478	-0.544	-0.493	-0.609	-0.489
(*)	0.149	0.281	0.25	0.249	0.242	0.301	0.249	0.29	0.26
Years French presence $\times$ 1800	0.118	-0.324	-0.25	-0.248	-0.235	-0.31	-0.241	-0.349	-0.252
(*)	0.122	0.249	0.223	0.227	0.218	0.268	0.223	0.269	0.235
Years French presence $\times$ 1850	0.072	-0.235	-0.159	-0.161	-0.161	-0.248	-0.157	-0.25	-0.157
(*)	0.204	0.275	0.254	0.252	0.246	0.29	0.251	0.314	0.264
Years French presence $\times$ 1875	-0.017	0.324	0.405	0.403	0.403	0.334	0.397	0.263	0.397
(*)	0.384	0.324	0.324	0.328	0.326	0.381	0.335	0.365	0.344
Years French presence $\times$ 1900	0.067	0.552	0.633	0.635	0.62	0.524	0.631	0.484	0.628
(*)	0.565	0.411	0.412	0.409	0.41	0.464	0.426	0.467	0.43
Observations	69	68	69	69	68	68	69	68	68
Number of states	12	12	12	12	12	12	12	12	12
p-value for joint significance after 1800	0.694	0.128	0.101	0.051	0.075	0.114	0.084	0.118	0.068

Table 8: Re-analysis (Leave-one-cluster-out) - (*) denotes sample variance